

5

Modules

60

Solved Q&A

13

Past Papers

40+

Key Formulas

2026

Latest Paper



Formula & Concept Cheatsheet

All key formulas across Modules 1-5 at a glance

Linear Regression

NORMAL EQUATION

$$\theta = (X^T X)^{-1} X^T y$$

MSE LOSS

$$J(\theta) = (1/2m) \sum (h\theta(x_i) - y_i)^2$$

GD UPDATE

$$\begin{aligned} \theta_j &:= \theta_j - \\ &\alpha \cdot (1/m) \sum (h\theta(x_i) - y_i) \cdot x_{ij} \end{aligned}$$

Regularization

RIDGE (L2)

$$J = \text{MSE} + \lambda \sum \theta_j^2 \rightarrow \text{shrinks all}$$

LASSO (L1)

$$J = \text{MSE} + \lambda \sum |\theta_j| \rightarrow \text{zeros some}$$

KEY DIFFERENCE

Lasso=sparse;
Ridge=all small
High-dim? $\rightarrow$ Use Lasso

Logistic Regression

SIGMOID

$$\sigma(z) = 1 / (1 + e^{-z}) \in (0, 1)$$

LOG-LOSS

$$J = -(1/m) \sum [y \log(\hat{h}) + (1-y) \log(1-\hat{h})]$$

GD UPDATE (SAME FORM!)

$$\begin{aligned} \theta_j &:= \theta_j - \\ &\alpha \cdot (1/m) \sum (\sigma(\theta^T x_i) - y_i) \cdot x_{ij} \end{aligned}$$

Decision Tree

ENTROPY

$$H(S) = -\sum p_i \log_2(p_i)$$

INFORMATION GAIN

$$\begin{aligned} IG(S, A) &= H(S) \\ &- \sum (|S_v| / |S|) H(S_v) \end{aligned}$$

GINI IMPURITY

$$\text{Gini} = 1 - \sum p_i^2$$

Evaluation Metrics

$$\begin{aligned} \text{Accuracy} &= (TP+TN) / (TP+TN+FP+FN) \\ \text{Precision} &= TP / (TP+FP) \\ \text{Recall} &= TP / (TP+FN) \\ F1 &= 2 \cdot P \cdot R / (P+R) \\ \text{RMSE} &= \sqrt{(\sum (y_i - \hat{y}_i)^2 / n)} \\ R^2 &= 1 - \text{SS}_{\text{res}} / \text{SS}_{\text{tot}} \end{aligned}$$

Normalization & Info Theory

MIN-MAX

$$x' = (x - \min) / (\max - \min)$$

Z-SCORE

$$x' = (x - \mu) / \sigma$$

ENTROPY / BITS

$$\begin{aligned} H &= -\sum p \log_2 p; \\ \text{bits} &= \lceil \log_2 N \rceil \end{aligned}$$

1

Module 1: Introduction to Machine Learning

Tom Mitchell's Definition · Types of ML · Learning System Design · Challenges

 Tom Mitchell's Definition

- **T (Task):** What the program does (classify, predict, cluster)
- **P (Performance):** How well (accuracy, RMSE, AUC-ROC)
- **E (Experience):** Training data it learns from
- A program *learns* when P on T improves with E

 Types of ML

- **Supervised:** Labeled data → Classification / Regression
- **Unsupervised:** No labels → Clustering, Dimensionality Reduction
- **Reinforcement:** Agent + Rewards → Sequential decisions
- **Semi-supervised:** Mix of labeled + unlabeled

 Learning System Design Steps

1. Choose training experience
2. Choose target function to learn
3. Choose representation for that function
4. Choose learning algorithm
5. Complete design & evaluate performance

 Challenges in ML

- Insufficient / noisy data
- Overfitting vs underfitting
- Feature engineering difficulty
- Interpretability vs accuracy
- Data leakage / distribution shift
- Scalability with large datasets

 Key Terminology

- **Feature:** Input variable / predictor
- **Label:** Target / output variable
- **Hypothesis:** A candidate learned function
- **Generalization:** Performance on unseen data
- **Inductive bias:** Model assumptions

 Discriminative vs Generative

- **Discriminative:** Models $P(y|x)$ directly → Logistic Regression, SVM
- **Generative:** Models $P(x|y)$ & $P(x,y)$ → Naïve Bayes, GMM
- Need density estimation / outlier detection → Use **Generative**

 **Exam Pattern:** Module 1 is always Q1. Expect 3–4 marks on identifying T/E/P from a scenario, and classifying the learning type. These questions repeat identically across years — master them fully.

 Previous Year Questions — Module 1

Q1. For a weather prediction model (Sunny/Windy/Rainy), identify E, T, and P in Tom Mitchell's definition. Name 2 performance measures P.

2023 MidSem Regular

2024 Makeup

3 Marks

 Repeats Every Year Answer

- **T:** Multi-class classification task — predict weather as Sunny, Windy, or Rainy
- **P:** Fraction of new instances correctly classified / probability of correct prediction on future dates
- **E:** Historical weather data (temperature, humidity, pressure + actual weather outcomes)

Performance Measures (any 2): Accuracy · Precision · Recall · F1-score · AUC-ROC · Misclassification Rate · Log-loss

Q2. For predicting tomorrow's rainfall in mm — which learning technique? What are 2 suitable performance measures?

2023 MidSem Regular

2 Marks

Regression

✓ Answer

Technique: Regression (Linear/Polynomial), because the target (mm of rainfall) is a *continuous* numeric value — not a discrete class.

Performance Measures (any 2):

- MSE (Mean Squared Error): $\sum(y_i - \hat{y}_i)^2/n$ — penalises large errors more
- RMSE (Root MSE): Same unit as target → easier to interpret
- MAE (Mean Absolute Error): $|y_i - \hat{y}_i|$ average — robust to outliers
- R^2 (Coefficient of determination): Proportion of variance explained

Q3. A Google ML system beat a human Go player. Version 1 learned from human gameplay; Version 2 learned by playing against itself with rewards. Describe this transition in ML approaches.

2023 MidSem Regular

2 Marks

Types of ML

✓ Answer

Transition from **Supervised Learning** → **Reinforcement Learning**.

- **V1 (Supervised):** Labeled human game moves as training data; learns to mimic expert gameplay.
- **V2 (Reinforcement):** No labeled data; learns by playing against itself, receiving win/lose rewards, discovering optimal strategy through trial and error.

Key distinction: Supervised needs human-labeled examples; RL discovers optimal behaviour autonomously via delayed reward signals.

Q4. Identify ML type for: (1) Self-driving car with feedback (2) Sentiment analysis on labeled reviews (3) Clustering unlabeled DNA sequences (4) Predicting insurance claim amount.

2024 MidSem Regular

2 Marks

★ Very Frequent

✓ Answer

- **Self-driving car:** Reinforcement Learning — learns via environment feedback (rewards)
- **Sentiment analysis (labeled):** Supervised — Classification (3 discrete classes)
- **DNA clustering (no labels):** Unsupervised — Clustering
- **Insurance claim amount:** Supervised — Regression (continuous numeric target)

Memory trick: Labels present? → Supervised. No labels? → Unsupervised. Environment rewards? → Reinforcement. Continuous output? → Regression. Discrete output? → Classification.

Q5. Recommend discriminative or generative classifier for: (a) classify job applications as good/bad, (b) detect lying applicants using density estimation / outlier detection.

2023 MidSem Regular 2 2 Marks Discriminative vs Generative

✓ Answer

Recommendation: Generative Classifier (e.g., Naïve Bayes, GMM)

- For **density estimation and outlier detection** you need $P(x|y)$ — the likelihood of features given a class. Only generative models compute this.
- Discriminative models (Logistic Regression, SVM) only compute $P(y|x)$ — the posterior class probability — and cannot model the data density needed for anomaly detection.

Rule: Generative models = $P(x|y) + P(y) \rightarrow$ full joint $P(x,y)$. Discriminative models = $P(y|x)$ only.

Q6. What is inductive bias? Why is it important in machine learning? Give 2 examples.

2021 EndSem 2 Marks Core Concept

✓ Answer

Inductive bias: The set of assumptions a learning algorithm makes to generalize from finite training examples to all possible inputs. Without it, a model cannot extrapolate beyond what it has seen.

Examples:

- Linear Regression: assumes a linear relationship between features and target
- Decision Tree (ID3): prefers shorter trees (Occam's Razor bias)
- KNN: assumes nearby points in feature space have similar labels

Why important: The No Free Lunch theorem shows no algorithm is universally best. Correct inductive bias matching the data structure is the key to good generalization.

Q7. A bank wants to predict loan default (yes/no) AND detect unusual spending patterns. What ML type for each? Justify.

ML Q&A PDF 2 Marks Application

✓ Answer

- **Loan default (yes/no):** Supervised — Binary Classification. Historical records with known outcomes serve as labeled training data.
- **Unusual spending patterns:** Unsupervised — Anomaly Detection / Clustering. No "unusual" label exists; the system learns the normal distribution and flags deviations.

Q8. Distinguish supervised, unsupervised, and reinforcement learning with one real-world example each.

Sample QP 3 Marks Comparison

✓ Answer

Aspect	Supervised	Unsupervised	Reinforcement
Training data	Labeled (X,y) pairs	Unlabeled (X only)	State-action-reward tuples
Goal	Learn $f: X \rightarrow y$	Discover hidden structure	Learn optimal policy π
Feedback	Labels (immediate)	None	Delayed reward
Example	Email spam detection	Customer segmentation	Robot navigation, AlphaGo

Q9. For the checkers-playing program example — define T, P, E. What kind of training experience is ideal and why?

2021 EndSem Regular 3 Marks Mitchell's Definition

✓ Answer

- **T:** Playing checkers (making legal moves to win the game)
- **P:** Win percentage against human and computer opponents
- **E:** Playing thousands of games against itself or opponents; observing board states and final outcomes

Ideal training experience: Direct feedback via self-play (Reinforcement Learning). Self-play generates diverse board states covering the full state space, unlike imitating human games which may miss many positions. The program learns the value function $V(\text{board}) \rightarrow \text{probability of winning from that state}$.

Q10. What are the key issues in designing a machine learning system? List at least 4 issues with brief explanations.

Sample QP / ML Q&A PDF 4 Marks Design Issues

✓ Answer

- **What training experience?** Direct vs indirect feedback, distribution coverage, human teacher vs self-exploration
- **What target function to learn?** Complete function vs evaluation function (affects what can be learned)
- **How to represent the function?** Linear, polynomial, neural network, decision tree — determines bias-variance tradeoff
- **How to learn from training data?** Gradient descent, closed-form solution, search-based methods
- **Overfitting:** Model memorizes noise $\rightarrow$ fails on unseen data. Use regularization, cross-validation.
- **Data quality:** Missing values, outliers, imbalanced classes degrade model performance
- **Scalability:** Algorithm must be efficient for large datasets ($O(n)$ vs $O(n^2)$ matters)
- **Interpretability vs accuracy:** Regulatory settings require explainable models even at cost of accuracy

Q11. A recommendation system learns which movies to suggest from user watch history and ratings. Identify T, E, P and the type of ML. Would you use a generative or discriminative approach?

Comprehensive Sample QP

3 Marks

Application

✓ **Answer**

- **T:** Recommending movies that a user will enjoy (ranking / rating prediction)
- **P:** User click-through rate / Mean Absolute Error of rating predictions / NDCG
- **E:** User watch history, ratings, demographic data
- **Type:** Supervised Learning (if using explicit ratings) OR Reinforcement Learning (if optimising long-term engagement)

Approach: Typically discriminative (e.g., collaborative filtering modelled as regression). Generative models (like Restricted Boltzmann Machines) can also be used for modelling user preference distributions.

 7 Types of Data Quality Issues

- **Missing values:** NULL/blank fields
- **Outliers:** Statistically extreme values (age=350)
- **Inconsistent format:** Dates in different formats
- **Duplicate records:** Same entity repeated
- **Invalid data type:** dev[at]mail vs dev@mail
- **Unit mismatch:** kg vs lbs in same column
- **Inconsistent naming:** AI vs Artificial Intelligence

 Data Preprocessing Steps

- **Imputation:** Fill missing (mean/median/mode)
- **Normalization/Scaling:** Min-max or Z-score
- **Encoding:** One-hot or label encoding
- **Outlier handling:** Remove, cap, or transform
- **Deduplication:** Remove exact duplicates
- **Feature selection:** Drop irrelevant features

 Train / Val / Test Split

- **Typical:** 70/15/15 or 80/10/10
- **Stratified split:** Preserve class proportions (key for imbalanced data)
- **Validation:** For hyperparameter tuning
- **Test:** NEVER touch until final evaluation
- **Data leakage:** Touching test during training → overly optimistic results

 Bias-Variance Tradeoff

- **High Bias (Underfitting):** Train $\approx$ Val, both HIGH → model too simple
- **High Variance (Overfitting):** Train LOW, Val HIGH → model too complex
- **Fix Bias:** More features, complex model, decrease λ
- **Fix Variance:** More data, regularisation, simpler model, increase λ

 Normalisation Formulas**MIN-MAX**

$$x' = (x - x_{\min}) / (x_{\max} - x_{\min}) \rightarrow [0, 1]$$

Z-SCORE (STANDARDISATION)

$$x' = (x - \mu) / \sigma \rightarrow \text{mean}=0, \text{std}=1$$

WHEN TO USE?

Min-Max: bounded range needed

Z-Score: Gaussian data, GD-based models

 Evaluation Metrics Insight

- **Accuracy misleads** with imbalanced classes
- **Precision:** Of predicted positives, how many are correct?
- **Recall/Sensitivity:** Of actual positives, how many did we catch?
- **F1:** Harmonic mean — balances precision & recall
- **AUC-ROC:** Model's discrimination ability
- **Lower threshold** → Higher recall, lower precision

 **Exam Pattern:** Module 2 appears in Q1b and Q2. Expect a data table with ≥ 6 issues to identify + fix. Also expect bias-variance diagnosis and a stratified split justification question. These repeat across 2023–2026.

 Previous Year Questions — Module 2

Q1. You are auditing a student academic dataset. Identify at least 6 data quality issues and suggest resolutions. [Table has: inconsistent dates, duplicate record, age=58 outlier, missing email, invalid email format dev[at]mail, inconsistent specialization AI vs Artificial Intelligence]

2024 MidSem Regular

4 Marks

★ Very Frequent

✓ **Answer — 6 Data Quality Issues**

1. **Inconsistent Date Formats** (15-Aug-2022, 2022-08-15, 15/08/2022) → Standardise to YYYY-MM-DD using a date parser
2. **Duplicate Record** (Rina appears twice with identical data) → Remove exact duplicates using deduplication on (name + email + specialization)
3. **Age Outlier** (Mansi = 58 for M.Tech student) → Replace with mean/median age, or flag for manual review (valid range: 20–35)
4. **Missing Email** (Shweta has blank email) → Flag as missing; contact institution or impute as "unknown"
5. **Invalid Email Format** (dev[at]mail.com instead of dev@mail.com) → Apply regex validation and correct or discard
6. **Inconsistent Specialization Naming** ("AI" vs "Artificial Intelligence") → Apply a mapping/lookup table to unify names

Pattern: Always look for (1) format mismatch, (2) numeric outliers, (3) missing values, (4) invalid data types, (5) exact duplicates, (6) naming inconsistencies.

Q2. Dataset: 10,000 loan records, only 800 (8%) approved. (a) How to split train/val/test? (b) Model A=95% accuracy, Model B=84% — which is more reliable? (c) Why keep test set untouched?

2024 MidSem Regular

5 Marks

★ Imbalanced + Metrics

✓ **Answer**

(a) **Stratified split:** Since only 8% are approved, use a *stratified 70/15/15* split to preserve the 8% approved ratio in all three sets. A random split risks having very few or zero positives in a subset.

(b) Despite higher accuracy, **Model A (95%) is likely misleading.** A model that predicts "rejected" for ALL 10,000 records achieves 92% accuracy — yet it is completely useless. Model A may be doing exactly this. Model B (84%) could be correctly identifying approved loans even if overall accuracy is lower.

→ Check precision, recall, F1-score, and ROC-AUC **specifically for the minority class** (approved loans).

(c) Using the test set during development causes *data leakage*: the model indirectly "sees" test patterns and reported performance will be overly optimistic. The test set simulates truly unseen real-world data.

Q3. "I doubled dataset size but validation RMSE stayed high while training RMSE kept dropping." (a) High bias or high variance? (b) One likely reason (c) One corrective action.

Dec 2025 MidSem

3 Marks

★ Bias-Variance

✓ **Answer**

(a) **Low Bias, High Variance = Overfitting.** Low training error + high validation error is the classic overfitting signature.

(b) **Likely reason:** Model is overly complex. Adding data helps reduce variance only if model size is fixed. If complexity grows with data, the gap persists. The doubled data may also have introduced more noise or a different distribution.

(c) **Corrective action:** Apply regularisation (Ridge L2 or Lasso L1) to penalise large weights and constrain model complexity. Other options: reduce polynomial degree, use dropout (NNs), or apply cross-validation with a complexity penalty.

Q4. Medical diagnostic: 1,000 patients, 150 have disease. System correctly identifies 120 diseased, incorrectly flags 85 healthy. (a) Confusion matrix (b) Accuracy, Precision, Recall, F1 (c) FN=\$50K, FP=\$2K — total cost? Threshold recommendation?

Jan 2026 Midsem Makeup

5 Marks

★ Metrics + Cost

✓ Answer

(a) Confusion Matrix:

	Pred Positive	Pred Negative
Actual Positive (150)	TP = 120	FN = 30
Actual Negative (850)	FP = 85	TN = 765

Accuracy = $(120+765)/1000 = 88.5\%$ Precision = $120/(120+85) = 120/205 = 58.5\%$ Recall = $120/(120+30) = 120/150 = 80.0\%$ F1-Score = $2 \times (0.585 \times 0.80) / (0.585 + 0.80) = 67.6\%$ FN cost = $30 \times \$50,000 = \$1,500,000$ ← dominant FP cost = $85 \times \$2,000 = \$170,000$ Total = $\$1,670,000$

Recommendation: Lower the classification threshold to increase recall (catch more diseased patients). FN cost is 25× higher than FP cost — we should prioritise reducing false negatives even at the expense of more false positives.

Key Insight: In medical contexts, missing a disease (FN) is far worse than an unnecessary test (FP). Lower threshold → higher recall → more FPs acceptable.

Q5. Ridge Regression: training error $\approx$ validation error $\approx$ high. High bias or high variance? What steps? Should you increase λ ?

2023 MidSem Regular 2

3 Marks

Bias-Variance + Regularisation

✓ Answer

Train $\approx$ Val AND both HIGH → **High Bias (Underfitting)**. Model too simple to capture the data pattern.

Steps:

- Increase model complexity (add polynomial/interaction features)
- **Decrease λ** (not increase!) — high λ over-constrains the model and worsens underfitting
- Add more informative features (feature engineering)
- Try a non-linear model

Critical: Do NOT increase λ when underfitting. Increasing λ increases bias further.

Q6. Normalise Total Runs [820,840,860,880,900] and Total Wickets [44,46,48,50,52] using Z-score normalisation. Show full calculation.

Dec 2025 MidSem 3 Marks Z-Score Calculation

✓ Answer

Total Runs: $\mu = (820+840+860+880+900)/5=860$, $\sigma^2 = [(-40)^2 + (-20)^2 + 0^2 + 20^2 + 40^2]/5=800$, $\sigma=28.28$ Total Wickets: $\mu=48$, $\sigma^2=[16+4+0+4+16]/5=8$, $\sigma=2.83$ Team A: Runs= $(820-860)/28.28=-1.41$, Wickets= $(44-48)/2.83=-1.41$ Team B: Runs= -0.71 , Wickets= -0.71 Team C: Runs= 0 , Wickets= 0 Team D: Runs= $+0.71$, Wickets= $+0.71$ Team E: Runs= $+1.41$, Wickets= $+1.41$

Q7. Models M1 and M2 at threshold $t=0.5$. M1: TP=3, FN=2, FP=1, TN=4. M2: TP=1, FN=4, FP=1, TN=4. Compute Precision, Recall, F1 for both.

Dec 2025 MidSem 5 Marks Metrics Calculation

✓ Answer

Model M1: TP=3, FN=2, FP=1, TN=4 Precision = $3/(3+1) = 0.75$ Recall = $3/(3+2) = 0.60$ F1 = $2 \times (0.75 \times 0.60) / (0.75 + 0.60) = 0.90/1.35 = 0.667$ Model M2: TP=1, FN=4, FP=1, TN=4 Precision = $1/(1+1) = 0.50$ Recall = $1/(1+4) = 0.20$ F1 = $2 \times (0.50 \times 0.20) / (0.50 + 0.20) = 0.20/0.70 = 0.286$

Conclusion: M1 is the better model across all three metrics.

Q8. What is overfitting? Key implications? How do regularisation and early stopping address it in logistic/linear regression?

2024 MidSem Regular 4 Marks Overfitting

✓ Answer

Overfitting: Model memorizes training noise; fails to generalise.

Implications: (1) Loss of generalisability, (2) Unreliable predictions on new data, (3) High model complexity.

Regularisation: Adds a penalty to the loss function discouraging large coefficients.

Ridge (L2): $J = \text{MSE} + \lambda \sum \theta_j^2 \rightarrow$ shrinks all weights; keeps all features Lasso (L1): $J = \text{MSE} + \lambda \sum |\theta_j| \rightarrow$ drives some weights to exactly 0 (feature selection)

Early Stopping: Monitor validation loss during training. When validation loss starts rising while training loss keeps falling $\rightarrow$ stop training. Prevents memorising noise.

Q9. 4 classes. (a) Minimum bits to encode a class label? (b) If uniform distribution, calculate entropy. (c) If non-uniform, does entropy increase or decrease? (d) Event A $p=0.2$ vs Event B $p=0.8$ — which needs more information?

2024 MidSem Regular

6 Marks

★ Information Theory

✓ Answer

(a) Minimum bits = $\lceil \log_2(4) \rceil = \lceil 2 \rceil = 2$ bits Why? 2 bits can represent 4 states: 00,01,10,11 — exactly enough for 4 classes. (b) Uniform $p = 1/4$ for each class: $H = -4 \times (1/4) \times \log_2(1/4) = -4 \times (1/4) \times (-2) = 2$ bits ← Maximum entropy equals $\log_2(N)$ for uniform distribution (c) Non-uniform → LOWER entropy. Entropy is maximised when all classes are equally likely. Any deviation from uniformity reduces entropy. Example: $p=[0.7, 0.1, 0.1, 0.1]$ $H = -(0.7 \times \log_2 0.7) - 3 \times (0.1 \times \log_2 0.1) \approx 0.36 + 0.996 = 1.36$ bits < 2 bits (d) Event A ($p=0.2$): $\text{info} = -\log_2(0.2) = 2.32$ bits ← needs MORE information Event B ($p=0.8$): $\text{info} = -\log_2(0.8) = 0.32$ bits Rare events are more surprising → require more bits to represent.

Q10. House price = $50 + 200 \times (\text{area}) + 5 \times (\text{age})$. (a) Interpret both coefficients. (b) What if area changes from m^2 to ft^2 ? (c) Which feature has stronger influence?

Dec 2025 MidSem

4 Marks

Coefficient Interpretation

✓ Answer

- **(a) Coefficient 200 (area):** Each additional m^2 adds ₹200 to the price, holding age constant.
- **Coefficient 5 (age):** Each additional year of age changes price by ₹5, holding area constant.
- (b) Changing units $\text{m}^2 \rightarrow \text{ft}^2$:** $1 \text{ m}^2 \approx 10.764 \text{ ft}^2$. Coefficient becomes $200/10.764 \approx 18.58$ per ft^2 . Model behaviour (predictions) stays identical — only the number changes.
- (c) Area has stronger influence.** Coefficient $200 \gg 5$. Each unit of area raises price $40\times$ more than a unit of age. For fair comparison across different-scale features, standardise inputs first.

Q11. Health dataset with TXN-ID, NAME, AGE, HEIGHT, WEIGHT, BLOOD GROUP, COVID-19 RESULT. Identify 5+ data quality issues from the sample.

2023 MidSem Regular 2

5 Marks

Data Quality

✓ Answer

1. **Invalid Name** (T010: SysUsr789) → Replace with correct name; apply alphanumeric-only validation
2. **Age Outlier** (T007: Age=350) → Impute with mean age; add range constraint [0–120]
3. **Missing Height** (T009: blank) → Impute with mean height
4. **Weight Unit Mismatch** (T008: 120lbs vs rest in kg) → Data transformation: 120 lbs = 54.4 kg; standardise units
5. **Invalid Blood Group** (T003: "Iam+ve") → Replace with NULL or correct value; apply domain-value validation
6. **COVID Result Encoding Mismatch** (T005=1, T009=0) → Standardise to Positive/Negative; apply data transformation

Q12. Explain K-fold cross-validation. What advantages does it have over a single train/test split?

ML Q&A PDF

3 Marks

Validation

✓ Answer

K-fold Cross-Validation:

1. Split dataset into k equal folds
2. For iteration $i = 1$ to k: train on $k-1$ folds, validate on fold i
3. Average the k validation scores for the final performance estimate

Advantages:

- Every sample is used for both training AND testing — no data waste
- Less sensitive to the specific split chosen (reduces sampling bias)
- Provides better variance estimate of model performance
- Essential for small datasets where held-out sets are costly

Common choice: $k=5$ or $k=10$. Leave-one-out ($k=n$) is the most thorough but computationally expensive.

Q13. High-order vs low-order model — compare in terms of bias, variance, training error, and validation error. When is a high-order model problematic?

Jan 2026 Makeup

3 Marks

Model Complexity

✓ Answer

Aspect	Low-Order (Simple)	High-Order (Complex)
Bias	High	Low
Variance	Low	High
Training error	High	Very low
Validation error	High (close to train)	Much higher than train
Problem	Underfitting	Overfitting

High-order model is problematic when: (1) training data is limited, (2) noise is high, (3) model memorises rather than learns patterns. Solutions: regularisation, reduce degree, more data, early stopping.

Linear Regression Basics

- Model: $h\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots = \theta^T x$
- Cost: $J(\theta) = (1/2m) \sum (h\theta(x_i) - y_i)^2$
- Goal:** Find θ that minimises $J(\theta)$
- Two methods:** Normal equation (direct) vs Gradient Descent (iterative)

Normal Equation (Direct Method)

$$\theta = (X^T X)^{-1} X^T y$$

- ✓ Closed-form exact solution
- ✓ No learning rate needed
- ✓ No iterations
- ✗ $O(n^3)$ — slow for large n features
- ✗ Fails if $X^T X$ is singular (use pseudo-inverse)

Gradient Descent (Iterative)

$$\begin{aligned} \theta_j &:= \theta_j - \alpha \cdot \partial J / \partial \theta_j \\ &= \theta_j - \alpha \cdot (1/m) \sum (h\theta(x_i) - y_i) \cdot x_{ij} \end{aligned}$$

Batch GD: all m samples per update
 Stochastic GD (SGD): 1 sample per update
 Mini-batch: k samples per update (best of both)

Regularisation — Ridge vs Lasso

Ridge (L2): $J = \text{MSE} + \lambda \sum \theta_j^2 \rightarrow$ Shrinks all weights towards 0 $\rightarrow$ Does NOT zero out any coefficient $\rightarrow$ Best for many small correlated features
 Lasso (L1): $J = \text{MSE} + \lambda \sum |\theta_j| \rightarrow$ Drives some weights to exactly 0 $\rightarrow$ Automatic feature selection $\rightarrow$ Best for high-dimensional sparse data
 Elastic Net: $\alpha \times \text{L1} + (1-\alpha) \times \text{L2}$ (combo)

Polynomial Basis Functions

- Add powers of features: $x, x^2, x^3, \dots$
- Still linear in parameters θ — uses same linear regression framework
- Allows fitting non-linear data patterns
- Higher degree $\rightarrow$ lower bias but higher variance
- Choose degree via cross-validation

When to Use Which Method

	Normal Eq.	Gradient Desc.
Features n	<10,000	>10,000
Speed	$O(n^3)$	$O(k \cdot m \cdot n)$
α needed?	No	Yes
Iterations	None	Many

Exam Pattern: Module 3 features L1/L2 regularisation comparison (choose which for high-dim data), gradient descent step calculations, and bias-variance analysis with polynomial degrees. A worked gradient descent update appears in most papers.

Previous Year Questions — Module 3

Q1. Describe how L1 (Lasso) and L2 (Ridge) regularisation alter the objective function. Which is more appropriate for a high-dimensional dataset with many potentially irrelevant predictors? Justify.

2024 MidSem Regular

4 Marks

★ Very Frequent

✓ Answer

Standard Linear Regression: $J(\theta) = (1/2m) \sum (h\theta(x_i) - y_i)^2$ Ridge (L2): $J(\theta) = \text{MSE} + \lambda \sum \theta_j^2 \rightarrow$ Adds squared magnitude penalty $\rightarrow$ Shrinks coefficients toward 0 but never exactly 0 $\rightarrow$ Controls complexity; all features retained Lasso (L1): $J(\theta) = \text{MSE} + \lambda \sum |\theta_j| \rightarrow$ Adds absolute value penalty $\rightarrow$ Drives some coefficients to EXACTLY 0 $\rightarrow$ Performs automatic feature selection

Answer for high-dimensional dataset: Lasso (L1) is more appropriate because:

- It eliminates irrelevant features by zeroing their coefficients
- Creates a sparse model (only important features retained)
- Enhances interpretability
- Prevents overfitting more aggressively when many features are noise

Rule: High-dim + irrelevant features $\rightarrow$ Lasso. Many correlated features + keep all $\rightarrow$ Ridge. Want both $\rightarrow$ Elastic Net.

Q2. You are the Lead ML Engineer building a regression model to predict Annual Default Risk. Lasso RMSE=0.081, Ensemble RMSE=0.062. Regulatory compliance requires auditability. (a) Which model? (b) Two disadvantages of the Ensemble in this setting.

Mar 2026 EndSem Regular

3 Marks

★ Lasso vs Ensemble

✓ Answer

(a) Recommended: Lasso Regression

- Encourages sparse solutions $\rightarrow$ automatically zeros irrelevant coefficients
- Linear structure supports auditability (single equation)
- Feature contribution is directly interpretable from coefficients
- Suitable when only 10–15 of 500 features are truly influential

(b) Two disadvantages of Ensemble in regulatory setting:

- **Low interpretability:** No single equation explaining predictions; impossible to trace feature-level contribution across hundreds of trees
- **Poor auditability:** Difficult to justify individual decisions to regulators; complex internal logic cannot be inspected or certified

Key principle: In regulated industries (banking, medical), explainability trumps raw accuracy. A slightly worse but interpretable model is preferred.

Q3. Compare Batch Gradient Descent, Stochastic Gradient Descent (SGD), and Mini-batch Gradient Descent. When would you use each?

ML Q&A PDF / Sample QP

4 Marks

Gradient Descent Variants

✓ Answer

Property	Batch GD	SGD	Mini-batch GD
Samples/update	All m	1	k (e.g. 32, 64)
Speed per epoch	Slow	Fast	Medium
Convergence	Smooth, stable	Noisy, oscillates	Balanced
Memory use	High	Low	Medium
Best for	Small datasets	Online learning, streaming	Large datasets, deep learning

Update rule (same form): $\theta_j := \theta_j - \alpha \cdot (1/\text{batch_size}) \cdot \sum (h\theta(x_i) - y_i) \cdot x_{ij}$

Q4. In gradient descent, $\partial J/\partial \theta_1$ is large while $\partial J/\partial \theta_2$ is near zero. What can you infer about the model parameters θ_1 and θ_2 ?

Dec 2025 MidSem

1 Mark

Gradient Interpretation

✓ Answer

- θ_1 : Large gradient means θ_1 is far from optimal. It needs significant adjustment (large step down the loss surface). Feature 1 contributes substantially to prediction error.
- θ_2 : Near-zero gradient means θ_2 has either reached its near-optimal value OR the corresponding feature (x_2) is largely irrelevant to the target. Very little update will occur for θ_2 .

Q5. One iteration of Batch Gradient Descent for logistic regression with $\theta = [0.1, 0.1, 0.1, 0.2]$, $\alpha = 0.1$, computed gradients: $\partial J/\partial \theta_0 = 0.065$, $\partial J/\partial \theta_1 = -0.1946$, $\partial J/\partial \theta_2 = 0.197$, $\partial J/\partial \theta_3 = 0.197$. Update all weights.

Dec 2025 MidSem

4 Marks

★ GD Calculation

✓ Answer

Update rule: $\theta_j := \theta_j - \alpha \cdot \partial J/\partial \theta_j$
 $\theta_0 = 0.1 - 0.1 \times 0.065 = 0.1 - 0.0065 = 0.0935$
 $\theta_1 = 0.1 - 0.1 \times (-0.1946) = 0.1 + 0.01946 = 0.1195$
 $\theta_2 = 0.1 - 0.1 \times 0.197 = 0.1 - 0.0197 = 0.0803$
 $\theta_3 = 0.2 - 0.1 \times 0.197 = 0.2 - 0.0197 = 0.1803$
Updated $\theta = [0.0935, 0.1195, 0.0803, 0.1803]$

Note: θ_1 increased because its gradient was negative (the parameter was below its optimal value). All others decreased slightly.

Q6. What is the Normal Equation for linear regression? Compare it with Gradient Descent — when to use each.

ML Q&A PDF

3 Marks

Normal Equation vs GD

✓ Answer

Normal Equation: $\theta = (X^T X)^{-1} X^T y$ Derivation: Set $\partial J / \partial \theta = 0 \rightarrow X^T (X\theta - y) = 0 \rightarrow X^T X\theta = X^T y \rightarrow \theta = (X^T X)^{-1} X^T y$

	Normal Equation	Gradient Descent
When to use	$n < 10,000$ features	$n > 10,000$ features
α (learning rate)	Not needed	Must be tuned
Iterations	None	Many needed
Time complexity	$O(n^3)$	$O(k \cdot m \cdot n)$
Feature scaling	Not necessary	Required for convergence

Q7. What is the bias-variance decomposition of expected prediction error? How does polynomial degree relate to bias and variance?

2021 EndSem

4 Marks

Bias-Variance Decomposition

✓ Answer

Expected Test Error = Bias² + Variance + Irreducible Noise
Bias²: Error from wrong assumptions (underfitting) High when model too simple (degree 1 for curved data)
Variance: Error from sensitivity to training data fluctuations (overfitting) High when model too complex (degree 10 for simple data)
Noise: Irreducible error from inherent randomness in data

Polynomial degree tradeoff:

- Degree 1 (linear): High bias, low variance → underfits curved data
- Degree 3–4: Sweet spot — low bias AND reasonable variance
- Degree 10+: Very low bias, very high variance → overfits, bad on test data

Use cross-validation to find the optimal degree.

Q8. What are the effects of learning rate α on gradient descent? What happens if α is too large? Too small?

ML Q&A PDF 2 Marks Gradient Descent

✓ Answer

- **Too large α :** Algorithm overshoots the minimum and may *diverge* (loss increases). J oscillates or grows. Never converges.
- **Too small α :** Algorithm takes very tiny steps $\rightarrow$ extremely slow convergence. May appear stuck. Computationally expensive.
- **Optimal α :** Loss decreases smoothly every iteration, converging to minimum efficiently.

Practical tip: Plot $J(\theta)$ vs iterations. Should decrease monotonically. If it oscillates $\rightarrow$ decrease α . If it decreases too slowly $\rightarrow$ increase α . Try $\alpha \in \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3\}$.

Q9. What is mean squared error (MSE) and why is it used as the cost function for linear regression? What are its limitations?

Sample QP 2 Marks Loss Function

✓ Answer

$J(\theta) = (1/2m) \sum_i (h_{\theta}(x_i) - y_i)^2$ The factor $1/2$ makes the derivative cleaner: $\partial J / \partial \theta_j = (1/m) \sum (h_{\theta}(x_i) - y_i) \cdot x_{i,j}$

Why MSE? It is convex (bowl-shaped) $\rightarrow$ guaranteed global minimum $\rightarrow$ gradient descent always converges. It is differentiable everywhere. It penalises large errors more (squared term).

Limitations:

- Sensitive to outliers (squared term amplifies large errors)
- Units are squared (use RMSE for same units as y)
- Not ideal for regression with heavy-tailed error distributions $\rightarrow$ use MAE or Huber loss instead

Q10. Linear Regression model shows Training RMSE = 12.4 and Test RMSE = 28.6. Diagnose the problem and suggest three remedies.

Jan 2026 Makeup 3 Marks Diagnosis

✓ Answer

Diagnosis: High Variance (Overfitting). Training RMSE much lower than Test RMSE — model memorised training data but does not generalise.

Three remedies:

1. **Apply Regularisation:** Ridge (L2) or Lasso (L1) to penalise large weights and constrain model complexity
2. **Reduce model complexity:** Lower polynomial degree; remove less important features
3. **Get more training data:** More diverse examples reduce variance
4. **Feature selection:** Remove redundant or irrelevant features that model is fitting to noise

Q11. What is R^2 (coefficient of determination)? Can it be negative? A model has $R^2 = 0.85$ on training data but $R^2 = 0.32$ on test data. What does this tell you?

ML Q&A PDF

3 Marks

R^2 Interpretation

✓ Answer

$R^2 = 1 - SS_{\text{res}} / SS_{\text{tot}}$ where: $SS_{\text{res}} = \sum (y_i - \hat{y}_i)^2$ (residual sum of squares) $SS_{\text{tot}} = \sum (y_i - \bar{y})^2$ (total sum of squares) $R^2 = 1$: Perfect fit. All variance explained. $R^2 = 0$: Model predicts just the mean — no better than a baseline. $R^2 < 0$: Model is WORSE than just predicting the mean. Yes, R^2 CAN be negative!

$R^2=0.85$ train, $R^2=0.32$ test interpretation:

- The large gap (0.85 vs 0.32) is a strong signal of **overfitting**.
- The model explains 85% of training variance but only 32% of test variance.
- Fix: regularisation, simpler model, more training data.

Q12. Linear Regression vs Logistic Regression — when would you use each? Can Linear Regression be used for binary classification? What is the problem with this?

Mar 2026 EndSem

3 Marks

★ Lin Reg vs Log Reg

✓ Answer

	Linear Regression	Logistic Regression
Output type	Continuous real number	Probability $\in (0,1)$
Use case	Regression (predicting a number)	Classification (predicting a class)
Loss function	MSE	Log-loss (cross-entropy)
Output range	$(-\infty, +\infty)$	$(0, 1)$ via sigmoid

Linear Regression for binary classification problem:

- Can be done by thresholding output at 0.5, but has serious problems
- Outputs can be >1 or <0 — not valid probabilities
- The linear decision boundary is less robust to outliers
- MSE loss is not well-suited for 0/1 labels (log-loss is better for classification)
- Logistic Regression's sigmoid output naturally provides calibrated probabilities

 From Regression to Classification

- Linear regression outputs unbounded values $(-\infty, +\infty)$
- Classification needs output in $(0,1)$ — a probability
- Solution: Pass linear score through **sigmoid function**
- Decision boundary: predict 1 if $P > 0.5$, else 0
- Boundary is where $\theta^T x = 0 \rightarrow$ linear in feature space

 Sigmoid Function

$\sigma(z) = 1 / (1 + e^{-z})$
 Properties: • $\sigma(0) = 0.5$ (boundary point) •
 $\sigma(z) \rightarrow 1$ as $z \rightarrow +\infty$ •
 $\sigma(z) \rightarrow 0$ as $z \rightarrow -\infty$ •
 Derivative: $\sigma'(z) = \sigma(z) \cdot (1 - \sigma(z))$ $z = \theta^T x$
 $= \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots$
 Predict 1 if $\sigma(z) \geq 0.5$ (i.e., $z \geq 0$)

 Log-Loss (Cross-Entropy)

$J(\theta) = - (1/m) \sum [y_i \log(\sigma(\theta^T x_i)) + (1 - y_i) \log(1 - \sigma(\theta^T x_i))]$
 Why not MSE for classification? MSE is non-convex with sigmoid $\rightarrow$ many local minima
 Log-loss is convex $\rightarrow$ unique global minimum guaranteed
 Log-loss gives large penalty for confident wrong predictions

 Gradient Descent Update

$\partial J / \partial \theta_j = (1/m) \sum (\sigma(\theta^T x_i) - y_i) \cdot x_{ij}$
 $\theta_j := \theta_j - \alpha \cdot (1/m) \sum (\sigma(\theta^T x_i) - y_i) \cdot x_{ij}$ ← Same mathematical form as Linear Regression! Key difference:
 $h\theta(x) = \sigma(\theta^T x)$ vs
 $h\theta(x) = \theta^T x$

 Log-Odds Interpretation

$\log[P(y=1|x) / P(y=0|x)] = \theta^T x$ • Each θ_j = log-odds ratio for feature j •
 Positive θ_j : increasing x_j increases $P(y=1)$ •
 Negative θ_j : increasing x_j decreases $P(y=1)$ •
 Large $|\theta_j| \neq$ "always class 1" — other features matter! •
 Overfitting: Large weights $\rightarrow$ steep sigmoid $\rightarrow$ overconfident

 Multiclass Classification

- One-vs-Rest (OvR)**: Train k binary classifiers, pick highest score
- One-vs-One (OvO)**: Train $k(k-1)/2$ classifiers
- Softmax regression**: Generalises logistic regression to k classes; outputs probability distribution over all classes; $\sum P(y=k|x) = 1$

 **Exam Pattern:** Module 4 is one of the highest-scoring modules. Common questions: sigmoid plot with increasing weights and overfitting explanation; batch GD update for logistic regression; log-odds interpretation; confusion matrix + precision/recall/F1. The IPL/loan prediction scenario recurs in 2025–2026 papers.

Q1. Plot the sigmoid function $\sigma(wx) = 1/(1+e^{-wx})$ for $w \in \{1, 10, 100\}$. Explain how changing weights leads logistic regression to overfit.

2024 MidSem Regular

4 Marks

★ Very Frequent

✓ Answer

Qualitative description of plots:

- **w=1**: Smooth, gradual S-curve. Transitions slowly from 0 to 1 over a wide range of x.
- **w=10**: Steeper S-curve. Transitions from 0 to 1 much more quickly around $x=0$.
- **w=100**: Near-vertical step function at $x=0$. Almost a hard threshold.

w=1: _____/----- (gradual) w=10: ____/----- (steeper) w=100: __/----- (near-step)

How large weights cause overfitting:

- High weights make the sigmoid *very steep* — even tiny changes in x produce extreme predictions (near 0 or 1).
- The model becomes *overconfident* — assigns probabilities close to 0 or 1 for even borderline cases.
- The decision boundary fits training data perfectly but is brittle — small input perturbations flip the prediction.
- This is classic overfitting: the model memorises the training boundary instead of learning a generalisable pattern.

Fix: L2 regularisation (adds $\lambda \sum \theta_j^2$ penalty) discourages large weights → smoother sigmoid → better generalisation.

Q2. Full logistic regression pipeline for IPL playoff prediction: Handle missing value for Team C (Mean Batsmen Played). Z-score normalise Total Runs and Wickets. Perform one batch GD iteration with $\theta=[0.1,0.1,0.1,0.2]$, $\alpha=0.1$. Calculate updated weights and cross-entropy loss.

Dec 2025 MidSem 10 Marks ★ Full Pipeline

✓ Answer

Step 1 — Handle missing value (Team C, Mean Batsmen):

Known values: 7, 5, 8, 4 $\rightarrow$ Mean = $(7+5+8+4)/4 = 24/4 = 6$ Imputed value for Team C = 6
(Mean imputation valid: numeric, continuous, approximately symmetric distribution)

Step 2 — Z-score normalisation:

Runs: $\mu=860$, $\sigma=28.28 \rightarrow$ Teams: [-1.41, -0.71, 0, +0.71, +1.41] Wickets: $\mu=48$, $\sigma=2.83 \rightarrow$ Teams: [-1.41, -0.71, 0, +0.71, +1.41]

Step 3 — Batch GD update:

Given: $\theta=[0.1, 0.1, 0.1, 0.2]$, $\alpha=0.1$, $m=5$ Computed gradients: $\partial J/\partial \theta_0=0.065$, $\partial J/\partial \theta_1=-0.1946$, $\partial J/\partial \theta_2=0.197$, $\partial J/\partial \theta_3=0.197$ Updated weights: $\theta_0 = 0.1 - 0.1 \times 0.065 = 0.0935$ $\theta_1 = 0.1 - 0.1 \times (-0.1946) = 0.1195$ $\theta_2 = 0.1 - 0.1 \times 0.197 = 0.0803$ $\theta_3 = 0.2 - 0.1 \times 0.197 = 0.1803$

Step 4 — Predicted probability for Team D:

$z = 0.0935 + 0.1195 \times (8) + 0.0803 \times (0.71) + 0.1803 \times (0.71) = 0.0935 + 0.956 + 0.057 + 0.128 = 1.234$ $\sigma(1.234) = 1/(1+e^{-1.234}) \approx 0.7746$ Team D has 77.46% probability of qualifying for playoffs.

Q3. A student claims "a large positive weight θ_{CGPA} guarantees that candidates with high CGPA are always selected." Is this correct? Justify using log-odds interpretation.

Jan 2026 Midsem Makeup 2 Marks Log-Odds Interpretation

✓ Answer

No, this conclusion is incorrect.

In logistic regression, the log-odds of selection is:

$$\log[P(\text{selected}) / (1-P(\text{selected}))] = \theta_0 + \theta_1 \times \text{Age} + \theta_2 \times \text{CGPA} + \theta_3 \times \text{IQ} + \theta_4 \times \text{Height}$$

- A large positive θ_{CGPA} means that increasing CGPA *increases* the log-odds (and thus probability) of selection, all else equal.
- But the final probability depends on ALL features simultaneously: $\theta_0 + \theta_{\text{Age}} \times \text{Age} + \theta_{\text{CGPA}} \times \text{CGPA} + \theta_{\text{IQ}} \times \text{IQ} + \dots$
- If other features (e.g., very low IQ, very high age) push the total score $\theta^T x$ very negative, even a high CGPA cannot guarantee selection.
- Additionally, $\sigma(z)$ is always in (0,1) — never exactly 1 — so no one is "always selected" with certainty.

Correct statement: A large positive θ_{CGPA} means CGPA has a strong positive influence on selection probability, but it never guarantees selection regardless of other features.

Q4. Why does logistic regression use log-loss instead of MSE? What happens to the MSE loss surface with a sigmoid activation?

ML Q&A PDF

3 Marks

Loss Function

✓ Answer

When MSE is combined with sigmoid activation: $J(\theta) = (1/2m)\sum(\sigma(\theta^T x) - y)^2$ — the resulting loss surface is **non-convex**. It has many local minima and saddle points, making gradient descent unreliable (may get stuck).

Log-loss (cross-entropy) is used because:

- When composed with sigmoid, it produces a *convex* loss surface → gradient descent is guaranteed to find the global minimum
- It provides a larger gradient for confidently wrong predictions → faster learning signal
- It is the Maximum Likelihood Estimation (MLE) objective for Bernoulli distribution — theoretically principled

Log-loss: $J = -(1/m)\sum[y \cdot \log(\hat{h}) + (1-y) \cdot \log(1-\hat{h})]$ If $y=1$ and $\hat{h} \approx 0$: penalty = $-\log(0) \rightarrow \infty$ (heavily penalises wrong confident prediction) If $y=1$ and $\hat{h} \approx 1$: penalty = $-\log(1) \rightarrow 0$ (no penalty for correct confident prediction)

Q5. Three models: Model A (Linear Regression + threshold), Model B (Logistic Regression), Model C (Decision Tree, depth=3). Observations: A outputs values outside [0,1]; B outputs near 0 or 1 for extremes; C shows sudden prediction changes for small input changes. (a) Why A goes outside [0,1] but B cannot. (b) Which overfits most with small data? (c) Which model's predictions change most abruptly?

Mar 2026 EndSem Regular

3 Marks

★ Model Comparison

✓ Answer

(a): Linear Regression computes $\hat{y} = \theta^T x$ — an unbounded linear function with range $(-\infty, +\infty)$. Logistic Regression applies sigmoid $\sigma(\theta^T x)$ which squashes all real inputs into $(0,1)$. The sigmoid never reaches exactly 0 or 1, making it impossible to output values outside that range.

(b): **Decision Tree** overfits most with small data. It learns piecewise-constant boundaries using axis-aligned splits — extremely flexible. With limited data, it can create a unique leaf for nearly every training point. Linear and Logistic Regression impose a global linear boundary (strong inductive bias), generalising better with fewer samples.

(c): **Decision Tree** (Model C) shows the most abrupt changes. A tiny shift in input can move a data point across a split threshold, completely changing the predicted class. This discontinuous behaviour is dangerous in high-stakes systems (banking, medical) because it lacks stability — two very similar individuals could receive opposite decisions.

Q6. What is the decision boundary in logistic regression? How does it change for a 2D problem with features x_1 and x_2 ?

ML Q&A PDF 2 Marks Decision Boundary

✓ Answer

The decision boundary is the set of points where the model is equally likely to predict either class, i.e., $\sigma(\theta^T x) = 0.5 \rightarrow \theta^T x = 0$.

For 2D (x_1, x_2) : $\theta_0 + \theta_1 x_1 + \theta_2 x_2 = 0 \rightarrow$ This is a straight LINE in 2D feature space $\rightarrow x_2 = -(\theta_0 + \theta_1 x_1) / \theta_2$. The line divides the feature space into two half-planes. Points on one side: predict class 1. Other side: predict class 0.

For polynomial features (e.g., add $x_1^2, x_1 x_2, x_2^2$): the boundary becomes a curve (ellipse, circle, parabola, etc.) — still linear in the *parameter space*.

Q7. What is the softmax function? How does it generalise logistic regression to K classes?

ML Q&A PDF 3 Marks Multiclass

✓ Answer

Softmax: $P(y=k|x) = \exp(\theta_k^T x) / \sum_j \exp(\theta_j^T x)$ Properties: • Output is a probability distribution over K classes • All values in $(0,1)$; sum = 1 • For K=2: reduces to logistic regression (sigmoid is special case of softmax) Loss function: Cross-entropy $= -\sum_k y_k \log(P(y=k|x))$

Aspect	Binary Logistic	Softmax (K-class)
Classes	2	$K \geq 2$
Output	Scalar $P(y=1 x)$	Vector $[P(y=1 x), \dots, P(y=K x)]$
Activation	Sigmoid $\sigma(z)$	Softmax $\exp(z_k) / \sum \exp(z_j)$
Parameters	1 weight vector θ	K weight vectors $\theta_1, \dots, \theta_K$

Q8. Logistic regression weights $w=[0.5, -0.3]$, $b=0.2$. New input $x=[2,1]$. (a) Compute $f(x)=w^T x+b$. (b) What class does the model predict? (c) Apply sigmoid to get probability.

Mar 2026 EndSem Regular 3 Marks Calculation

✓ Answer

(a) $f(x) = w^T x + b = (0.5 \times 2) + (-0.3 \times 1) + 0.2 = 1.0 - 0.3 + 0.2 = 0.9$ (b) Since $f(x) = 0.9 > 0$, $\text{sigmoid}(0.9) > 0.5 \rightarrow$ Predict positive class (Class 1 / Default) (c) $P(y=1|x) = \sigma(0.9) = 1/(1+e^{-0.9}) = 1/(1+0.4066) = 1/1.4066 \approx 0.711$ Interpretation: 71.1% probability of default.

Q9. What is Precision-Recall tradeoff? When would you prioritise Precision over Recall and vice versa? Give concrete examples.

Sample QP 3 Marks Metrics

✓ Answer

As threshold $\uparrow$: Fewer positive predictions $\rightarrow \uparrow$ Precision, $\downarrow$ Recall
As threshold $\downarrow$: More positive predictions $\rightarrow \downarrow$ Precision, $\uparrow$ Recall

Prioritise Recall (sensitivity) when:

- Missing a positive is costly: cancer screening (missing disease = death), fraud detection (missing fraud = financial loss)
- FN cost $\gg$ FP cost $\rightarrow$ lower threshold to catch more positives

Prioritise Precision when:

- False alarms are costly: spam filter (blocking legitimate email), drug recommendation (giving wrong medication)
- FP cost $\gg$ FN cost $\rightarrow$ higher threshold to minimise false positives

Use F1-score when you want a single metric balancing both. Use F-beta score ($\beta > 1$ weights recall more, $\beta < 1$ weights precision more).

Q10. What are discriminant functions in linear classification? What is the role of the bias term θ_0 ?

ML Q&A PDF 2 Marks Discriminant Functions

✓ Answer

A **discriminant function** computes a score for each class; the class with the highest score is predicted: $\hat{y} = \operatorname{argmax}_k f_k(x)$.

For binary classification: $f(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots = \theta^T x$. Positive score $\rightarrow$ class 1; negative $\rightarrow$ class 0.

Role of bias term θ_0 : Shifts the decision boundary away from the origin. Without it, the decision boundary is forced to pass through the origin, severely restricting which datasets can be classified correctly. θ_0 controls the position of the boundary, while $\theta_1, \theta_2, \dots$ control its orientation.

Q11. An instructor critiques a student who claims "doubling dataset size doesn't help since validation RMSE stayed high". How would you respond using model complexity and bias-variance concepts?

Dec 2025 MidSem 3 Marks Bias-Variance + Critique

✓ Answer

The student's conclusion is **incorrect**. More data generally helps reduce variance, but the observation reveals a more nuanced issue:

- The pattern (low train error, high val error) → high variance / overfitting
- Adding data *should* reduce variance, but only if model complexity is held constant
- If the model complexity grew with the data (e.g., more polynomial features were added), the variance gap can persist
- Also: if the doubled data introduced distribution shift or more noise, overfitting can worsen

Corrective actions: Apply regularisation (not just add data); reduce model complexity; use cross-validation; normalise inputs.

Model is **Low Bias, High Variance**. Adding data is necessary but not sufficient if the model is too complex.

Q12. What is regularisation in logistic regression? How does L2 regularisation affect the decision boundary? Compare L1 vs L2 for logistic regression.

ML Q&A PDF 3 Marks Regularisation

✓ Answer

Regularised log-loss: $J(\theta) = -\frac{1}{m} \sum [y \cdot \log(\sigma(\theta^T x)) + (1-y) \cdot \log(1-\sigma(\theta^T x))] +$
 $(\lambda/2m) \sum \theta_j^2 \leftarrow \text{L2 term OR } + (\lambda/m) \sum |\theta_j| \leftarrow \text{L1 term}$

Effect of L2 on decision boundary: Penalising large weights prevents the sigmoid from becoming very steep → smoother, less extreme boundary → reduces overconfidence → better generalisation.

Feature	L2 (Ridge)	L1 (Lasso)
Effect on weights	Shrinks all weights small	Sets some weights to exactly 0
Feature selection	No (keeps all features)	Yes (automatic selection)
Solution	Unique (smooth)	Non-unique (sparse)
Best for	Correlated features	High-dim, sparse data

Q13. A binary classifier achieves 95% accuracy on a dataset where 95% of samples are class 0. Is this a good model? What would you use instead of accuracy?

[Sample QP / ML Q&A PDF](#)

2 Marks

Class Imbalance

✓ Answer

No, 95% accuracy is NOT a good model here.

A *trivial classifier* that predicts class 0 for every input achieves 95% accuracy — yet it never detects class 1. This is the accuracy paradox.

Better metrics for imbalanced data:

- **Precision:** Of all predicted positives, how many are truly positive?
- **Recall:** Of all actual positives, how many did we detect?
- **F1-score:** Harmonic mean of precision & recall
- **ROC-AUC:** Area under the ROC curve — measures discrimination ability across all thresholds
- **PR-AUC:** Area under Precision-Recall curve — better for highly imbalanced data
- **Cohen's Kappa:** Measures agreement beyond chance

🌳 What is a Decision Tree?

- A flowchart of yes/no questions to classify data
- Each **internal node**: a feature to test
- Each **branch**: an outcome of the test
- Each **leaf node**: a class label
- Built top-down, greedily, recursively
- **Fully interpretable** — can be read as "if-then" rules

📊 Entropy & Information Gain

Entropy: $H(S) = -\sum_i p_i \log_2(p_i)$ • $H=0$: pure node (all same class)

- $H=1$: perfectly mixed (50/50 binary)
- H is maximised at uniform distribution

Info Gain: $IG(S,A) = H(S) - \sum_v (|S_v|/|S|) \cdot H(S_v)$ • Choose attribute A with MAXIMUM IG • IG = how much entropy is reduced by splitting on A

🔍 ID3 Algorithm

- If all examples same class → return leaf with that class
- If no attributes left → return leaf with majority class
- Otherwise: compute IG for all attributes
- Select attribute A^* with highest IG as root
- For each value v of A^* : create branch, recurse

ID3 stops when: pure leaf OR no attributes OR no examples

🔍 Gini Impurity (Alternative)

Gini(S) = $1 - \sum_i p_i^2$ • Gini=0: pure node (one class only) • Gini=0.5: perfectly mixed (binary, 50/50)

- Faster to compute than entropy (no log)
- Used in CART algorithm (scikit-learn default)

Gini Gain = $Gini(S) - \sum_v (|S_v|/|S|) \cdot Gini(S_v)$

⚠️ Overfitting & Pruning

- **Problem:** Deep trees memorise training data → poor test performance
- **Pre-pruning:** Stop early if IG below threshold, or max depth reached, or too few samples in node
- **Post-pruning:** Grow full tree then prune branches that don't improve validation accuracy
- **MDL principle:** Prefer smaller trees that explain data well

🔍 Handling Special Cases

- **Continuous attributes:** Sort values, try each midpoint as threshold. Choose best threshold per attribute.
- **Missing attributes:** Use most common value, or use probabilistic assignment across branches
- **Multi-valued attributes:** Gain Ratio = $IG / \text{Split_Info}$ to avoid bias toward high-cardinality attributes (C4.5)

🔍 **Exam Pattern:** Module 5 is heavily tested in both midsem and endsem. Questions always involve: (1) calculating entropy of the full dataset, (2) calculating entropy of a branch, (3) computing information gain for one attribute, (4) determining if a branch is pure and what attribute to split on next. Know the full tennis/purchasing dataset example thoroughly.

📄 Previous Year Questions — Module 5

Q1. Customer purchasing dataset (14 records, attributes: Agegroup, Income Level, Occupation, Target: Purchased Yes/No). (i) Calculate entropy of entire dataset: 9 Yes, 5 No. (ii) Choose Agegroup as root — calculate entropy of "Young" branch (3 Yes, 1 No). Is it pure? Which attribute to split on next? (iii) Compare Occupation vs Income Level — which is better without full calculation? (iv) Classify new customer {Young, Low, Professional}.

Dec 2025 MidSem 8 Marks ★ Very Frequent — Full Tree Build

✓ Answer

(i) Entropy of full dataset (9 Yes, 5 No, total 14):

$$p(\text{Yes}) = 9/14 \approx 0.643 \quad p(\text{No}) = 5/14 \approx 0.357 \quad H(S) = -(9/14)\log_2(9/14) - (5/14)\log_2(5/14) = -0.643 \times (-0.637) - 0.357 \times (-1.485) = 0.409 + 0.530 \approx 0.940 \text{ bits}$$

(ii) Young branch (3 Yes, 1 No, total 4):

$$H(\text{Young}) = -(3/4)\log_2(3/4) - (1/4)\log_2(1/4) = -0.75 \times (-0.415) - 0.25 \times (-2.0) = 0.311 + 0.500 \approx 0.811 \text{ bits}$$

Is it pure? NO (entropy = 0.811 $\neq$ 0; contains both Yes and No).
 Next attribute to split on: Income Level
 Income=Low: 1 record $\rightarrow$ No (pure leaf: No)
 Income=Medium: 2 records $\rightarrow$ Yes, Yes (pure leaf: Yes)
 Income=High: 1 record $\rightarrow$ Yes (pure leaf: Yes) $\rightarrow$ Income Level perfectly separates the Young subset!

(iii) Income Level vs Occupation comparison:

- **Income Level:** Low $\rightarrow$ all No (pure); Medium $\rightarrow$ all Yes (pure); High $\rightarrow$ all Yes (pure). **Perfectly separates classes $\rightarrow$ maximum IG.**
- **Occupation:** Student $\rightarrow$ mix of Yes and No (impure); Professional $\rightarrow$ mix of Yes and No (impure); Retired $\rightarrow$ mix. High entropy in all branches $\rightarrow$ very low IG.
- **Conclusion: Income Level is the better attribute.**

(iv) Classify {Young, Low, Professional}:

Young branch has 3 Yes, 1 No. Majority class = Yes. $\rightarrow$ Predicted class: **Yes (will purchase).**

Q2. Loan default dataset (14 samples: 9 Repay, 5 Default). Income Level: High $\rightarrow$ 6 Repay, 2 Default; Low $\rightarrow$ 3 Repay, 3 Default. Using entropy-based IG, determine whether Income Level is a good split attribute. Show all steps.

Jan 2026 Midsem Makeup 6 Marks ★ Full IG Calculation

✓ Answer

Step 1: Parent entropy (9 Repay, 5 Default, total 14) $H(S) = -(9/14)\log_2(9/14) - (5/14)\log_2(5/14) \approx 0.940$ bits
 Step 2: Child node entropies
 Income = High (6 Repay, 2 Default, total 8): $H(\text{High}) = -(6/8)\log_2(6/8) - (2/8)\log_2(2/8) = -0.75 \times (-0.415) - 0.25 \times (-2.0) = 0.311 + 0.500 \approx 0.811$ bits
 Income = Low (3 Repay, 3 Default, total 6): $H(\text{Low}) = -(3/6)\log_2(3/6) - (3/6)\log_2(3/6) = -0.5 \times (-1.0) - 0.5 \times (-1.0) = 0.5 + 0.5 = 1.0$ bits (max entropy, perfectly mixed)
 Step 3: Weighted entropy after split $H_{\text{after}} = (8/14) \times H(\text{High}) + (6/14) \times H(\text{Low}) = (8/14) \times 0.811 + (6/14) \times 1.0 = 0.4634 + 0.4286 = 0.892$ bits
 Step 4: Information Gain $IG = H(S) - H_{\text{after}} = 0.940 - 0.892 \approx 0.048$ bits
 Conclusion: $IG \approx 0.048$ is very LOW. Income Level is NOT a good split attribute for this dataset — it provides minimal reduction in entropy.

Q3. The classic Tennis dataset: Outlook (Sunny/Overcast/Rain), Humidity (High/Normal), Wind (Weak/Strong). Describe how to build the decision tree using ID3. Which attribute is selected first and why?

2021 EndSem / ML Q&A PDF

6 Marks

★ Classic Example

✓ Answer

Dataset: 14 examples, 9 Play=Yes, 5 Play=No $H(S) = 0.940$ bits Information Gains (given/computed): $IG(\text{Outlook}) \approx 0.246$ bits ← HIGHEST → selected as root $IG(\text{Humidity}) \approx 0.151$ bits $IG(\text{Wind}) \approx 0.048$ bits $IG(\text{Temperature}) \approx 0.029$ bits

Tree after selecting Outlook as root:

- **Outlook = Overcast:** All 4 examples are Yes → Leaf: Play
- **Outlook = Sunny:** 2 Yes, 3 No → Not pure; further split on Humidity → Humidity=High: No; Humidity=Normal: Yes
- **Outlook = Rain:** 3 Yes, 2 No → Not pure; further split on Wind → Wind=Strong: No; Wind=Weak: Yes

Final rules:

- Overcast → Play
- Sunny AND Humidity=Normal → Play
- Sunny AND Humidity=High → Don't Play
- Rain AND Wind=Weak → Play
- Rain AND Wind=Strong → Don't Play

Q4. What is the MDL (Minimum Description Length) principle? How does it apply to decision tree learning?

2021 EndSem Makeup / ML Q&A PDF

3 Marks

MDL Principle

✓ Answer

MDL Principle: The best model is the one that provides the shortest total description of the data. It is a formalisation of Occam's Razor using information theory.

Total description length = $L(\text{model}) + L(\text{data} \mid \text{model})$ $L(\text{model})$: bits needed to describe the hypothesis (tree structure + labels) $L(\text{data} \mid \text{model})$: bits needed to encode exceptions/errors with that model MDL goal: minimise $L(\text{model}) + L(\text{data} \mid \text{model})$

Application to Decision Trees:

- A complex tree (many nodes) has a long $L(\text{model})$ but short $L(\text{data} \mid \text{model})$ — fits data well
- A simple tree has a short $L(\text{model})$ but longer $L(\text{data} \mid \text{model})$ — has more errors
- MDL finds the sweet spot that prevents overfitting
- Equivalent to applying a regularisation penalty proportional to tree size
- Used in post-pruning: prune branches where the gain in $L(\text{data} \mid \text{model})$ doesn't justify the cost in $L(\text{model})$

Q5. How does ID3 handle continuous-valued attributes (e.g., Temperature in degrees)? Describe the threshold selection process.

2021 EndSem Regular

3 Marks

Continuous Attributes

✓ Answer

ID3 discretises continuous attributes dynamically during tree construction:

1. **Sort** all examples by the continuous attribute value (e.g., temperature: 60, 70, 80, 90 °F)
2. **Find candidate thresholds:** Midpoints between consecutive values where the class label changes (60.5, 75, 85)
3. **Compute IG for each threshold:** For each threshold t , create a binary split ($x \leq t$) vs ($x > t$). Compute IG as if it were a binary attribute.
4. **Select the best threshold:** The t that maximises IG for this attribute
5. **Compare** this best IG against other attributes and select overall best

Note: The same continuous attribute can be used multiple times along a branch with different thresholds. Discrete attributes can only be used once.

Q6. What is the problem with using Information Gain alone to select split attributes? How does Gain Ratio (C4.5) fix this?

ML Q&A PDF

3 Marks

IG Bias & Gain Ratio

✓ Answer

Problem with raw IG: IG is biased towards attributes with many distinct values. An attribute like "Transaction ID" (unique for every row) would have $IG \approx H(S)$ — perfectly splitting the data into pure singletons — but is completely useless for prediction on new data.

Gain Ratio (C4.5): $GainRatio(S, A) = IG(S, A) / SplitInfo(S, A)$ $SplitInfo(S, A) = -\sum_v (|S_v|/|S|) \cdot \log_2(|S_v|/|S|)$ (This is the entropy of the attribute itself) Effect: Penalises attributes that create many small branches. A high-cardinality attribute has high SplitInfo $\rightarrow$ lower GainRatio $\rightarrow$ less likely selected.

ID3 uses IG. C4.5 uses Gain Ratio. CART uses Gini Impurity.

Q7. Compare Decision Trees with Logistic Regression on: interpretability, decision boundary shape, overfitting tendency, sensitivity to outliers.

Mar 2026 EndSem Regular

3 Marks

★ Model Comparison

✓ Answer

Property	Decision Tree	Logistic Regression
Interpretability	Very high (if-then rules)	Medium (coefficients)
Decision boundary	Axis-aligned, piecewise constant	Smooth linear hyperplane
Overfitting risk	High (deep trees)	Lower (especially with regularisation)
Sensitivity to outliers	Moderate	High (outliers shift boundary)
Feature scaling needed	No	Yes
Handles non-linearity	Naturally	Only with feature engineering
Probabilistic output	No (class labels)	Yes (calibrated probabilities)
Prediction stability	Low (abrupt changes)	High (smooth function)

Q8. A decision tree is built on training data and achieves 100% training accuracy but 62% test accuracy. Diagnose and propose solutions.

ML Q&A PDF

3 Marks

Overfitting

✓ Answer

Diagnosis: Severe Overfitting. 100% training accuracy with 62% test accuracy — the tree has memorised every training example (likely one leaf per example or very deep).

Solutions:

1. **Pre-pruning (restrict tree growth):** Set max_depth (e.g., 5); minimum samples per leaf (e.g., 5); minimum IG threshold for splitting
2. **Post-pruning:** Grow full tree then prune back branches that don't improve validation accuracy (reduced error pruning)
3. **Ensemble methods:** Use Random Forest (many trees + feature randomness) or Gradient Boosting to reduce variance
4. **Get more training data:** More data forces the tree to generalise rather than memorise

Q9. Calculate entropy for a node with 6 class-A and 6 class-B examples. Now split on an attribute giving: Branch1 = 5A + 1B, Branch2 = 1A + 5B. Calculate IG for this split.

ML Q&A PDF / Sample QP

3 Marks

Entropy + IG Calculation

✓ Answer

Parent node (6A, 6B, total 12): $H(S) = -(6/12)\log_2(6/12) - (6/12)\log_2(6/12) = -0.5 \times (-1) - 0.5 \times (-1) = 1.0$ bits (maximum entropy) Branch 1 (5A, 1B, total 6): $H(B1) = -(5/6)\log_2(5/6) - (1/6)\log_2(1/6) = -0.833 \times (-0.263) - 0.167 \times (-2.585) = 0.219 + 0.431 \approx 0.650$ bits Branch 2 (1A, 5B, total 6): $H(B2) = -(1/6)\log_2(1/6) - (5/6)\log_2(5/6) =$ same as $H(B1) \approx 0.650$ bits (symmetric) Weighted entropy after split: $H_{\text{after}} = (6/12) \times 0.650 + (6/12) \times 0.650 = 0.650$ bits Information Gain: $IG = H(S) - H_{\text{after}} = 1.0 - 0.650 = 0.350$ bits The split reduces entropy by 0.35 bits – a good split!

Q10. What are the advantages and disadvantages of Decision Trees? When should you NOT use a decision tree?

Sample QP / ML Q&A PDF

4 Marks

Pros & Cons

✓ Answer

Advantages ✓

Extremely interpretable (white-box model)

Handles both numeric and categorical features

No feature scaling required

Non-linear decision boundary naturally

Fast inference (follow branches)

Handles missing values

Disadvantages ✗

High variance — small data changes → very different tree

Overfits easily without pruning

Biased toward attributes with many values (raw IG)

Cannot represent XOR-type relationships well

Axis-aligned splits — inefficient for diagonal boundaries

Greedy construction → may miss globally optimal tree

When NOT to use: When the relationship between features and target is roughly linear (use logistic/linear regression instead). When features are continuous with complex correlations (SVMs or neural networks are better). When stability is critical (small input changes → same output).

Q11. What is the difference between ID3, C4.5, and CART algorithms for building decision trees?

[ML Q&A PDF](#)

3 Marks

[Algorithm Comparison](#)

✓ Answer

Feature	ID3	C4.5	CART
Split criterion	Information Gain	Gain Ratio	Gini Impurity
Attribute types	Categorical only	Categorical + Continuous	Categorical + Continuous
Missing values	No handling	Handles via probabilities	Handles via surrogates
Pruning	No (grows full tree)	Post-pruning (pessimistic)	Cost-complexity pruning
Tree structure	Multi-way splits	Multi-way splits	Binary splits only
Task	Classification only	Classification only	Classification AND Regression
Used in	Teaching/theory	Weka	scikit-learn



Module Formula Tree — Visual Study Guide

Click any branch to expand · Memory tricks in tags · High-exam-value in

AIMLCZG565 Machine Learning — Modules 1–5 Formula & Concept Tree ▶

Module 1 — Introduction to Machine Learning ▶

★ Tom Mitchell's Definition: T / E / P ▶

T = Task (what the program does: classify, predict, cluster)

P = Performance measure (accuracy, RMSE, AUC, F1)

E = Experience = Training Data

Trick: "TPE" — Think "The Program Experiences"

Exam: Weather prediction → T=classification, P=accuracy, E=historical weather data

★ Types of Machine Learning ▶

Supervised: Labels present → Classification (discrete) / Regression (continuous)

Unsupervised: No labels → Clustering, Dimensionality Reduction, Anomaly Detection

Reinforcement: Agent + Reward signals → learns optimal policy π

Trick: "Labels? → Supervised. No Labels? → Unsupervised. Rewards? → Reinforcement."

Trick: "Continuous output → Regression. Discrete output → Classification."

Exam trap: Self-driving car = Reinforcement (not supervised, even if it learns from humans)

Discriminative vs Generative Classifiers ▶

Discriminative: Models $P(y|x)$ directly → Logistic Reg., SVM, Decision Tree

Generative: Models $P(x|y) + P(y) = P(x,y)$ → Naïve Bayes, GMM

Trick: Need density estimation / outlier detection? → Always Generative (needs $P(x|y)$)

Inductive Bias & Learning System Design ▶

Inductive bias = assumptions a model makes to generalise beyond training data

No Free Lunch theorem: No single algorithm is best for all problems

Design steps: (1) Training experience (2) Target function (3) Representation (4) Algorithm (5) Evaluate

Trick: "TREAT" — Training, Representation, Evaluation, Algorithm, Task

Module 2 — ML Workflow & Data Preprocessing ▶

★ Data Quality Issues (7 Types)

1. Missing values → Mean/Median/Mode imputation
2. Outliers → Remove, cap, or replace with mean (age=350 → use mean age)
3. Inconsistent format → Standardise (dates: 15-Aug → YYYY-MM-DD)
4. Duplicate records → Deduplication on key columns
5. Invalid data type → Regex validation (dev[at]mail → dev@mail)
6. Unit mismatch → Transform all to same unit (lbs → kg)
7. Inconsistent naming → Mapping table (AI vs Artificial Intelligence)

💡 Trick: "MODI-UV" — Missing, Outliers, Duplicates, Inconsistent format, Unit mismatch, Validation (type), Version (naming)

★ Normalisation Formulas

Min-Max: $x' = (x - x_{\min}) / (x_{\max} - x_{\min}) \rightarrow \text{range } [0, 1]$

Z-Score: $x' = (x - \mu) / \sigma \rightarrow \text{mean}=0, \text{std}=1$

💡 Trick: "Min-Max → BOUNDED box [0,1]. Z-Score → ZERO mean (Gaussian-friendly)."

✅ Exam: Z-score for gradient descent models. Min-max when bounded range required.

★ Bias-Variance Tradeoff

Total Error = Bias² + Variance + Irreducible Noise

High Bias (Underfitting): Train HIGH ≈ Val HIGH → model too simple

High Variance (Overfitting): Train LOW, Val HIGH → model too complex

Fix High Bias: MORE complexity, DECREASE λ , add features

Fix High Variance: MORE data, regularisation, INCREASE λ , simpler model

💡 Trick: "Both errors high = Underfitting (bias). Gap between train/val = Overfitting (variance)."

💡 Trick: " λ up → Bias up, Variance down. λ down → Bias down, Variance up."

★ Evaluation Metrics Formulas

Accuracy = $(TP + TN) / (TP + TN + FP + FN)$

Precision = $TP / (TP + FP)$ ← "Of predicted +, how many truly +"

Recall = $TP / (TP + FN)$ ← "Of actual +, how many caught"

F1-score = $2 \times P \times R / (P + R)$ ← harmonic mean

RMSE = $\sqrt{(\sum (y_i - \hat{y}_i)^2 / n)}$ $R^2 = 1 - SS_{\text{res}}/SS_{\text{tot}}$

💡 Trick: "Precision = Positive predictive value. Recall = Sensitivity = True Positive Rate."

🧠 Trick: "Accuracy lies on imbalanced data. Always check F1 + AUC-ROC."

✅ Exam: FN cost >> FP cost (medical) → Lower threshold → Higher Recall. FP cost >> FN (spam) → Higher Precision.

Train / Val / Test Split

Typical split: 70 / 15 / 15 or 80 / 10 / 10

Stratified split: Preserve class proportions — REQUIRED for imbalanced datasets

Data Leakage: Using test data during training → overly optimistic performance

🧠 Trick: "Test set is a time capsule — sealed until final evaluation."

Module 3 — Linear Regression

★ The Model & Cost Function

Model: $h\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots = \theta^T x$

Cost (MSE): $J(\theta) = (1/2m) \sum_i (h\theta(x_i) - y_i)^2$

Why 1/2? → Makes derivative clean: $\partial J / \partial \theta_j = (1/m) \sum (h\theta(x_i) - y_i) \cdot x_{ij}$

🧠 Trick: MSE is CONVEX for linear models → gradient descent always finds global minimum.

★ Normal Equation (Direct Solution)

$\theta = (X^T X)^{-1} X^T y$

✓ No iterations, no learning rate, exact solution

✗ $O(n^3)$ — too slow when features $n > 10,000$

🧠 Trick: "Normal Eq = one-shot formula. Gradient Descent = roll-downhill loop."

✅ $n < 10,000$ features → Normal Equation. $n > 10,000$ → Gradient Descent.

★ Gradient Descent (Iterative)

Update: $\theta_j := \theta_j - \alpha \cdot (1/m) \sum (h\theta(x_i) - y_i) \cdot x_{ij}$

Batch GD: all m samples per update → stable, slow

SGD: 1 sample per update → fast, noisy

Mini-batch: k samples → best of both (used in deep learning)

🧠 Trick: " α too big → DIVERGE (bounce). α too small → CRAWL (slow). Plot J vs iter to check."

🧠 Trick: "Large gradient → far from optimal. Near-zero gradient → close to optimal or irrelevant feature."

★ Regularisation — Ridge vs Lasso

Ridge (L2): $J = \text{MSE} + \lambda \sum \theta_j^2$ → shrinks all weights, keeps all features

Lasso (L1): $J = \text{MSE} + \lambda \sum |\theta_j|$ → drives some weights to EXACTLY zero

Elastic Net: $\alpha \cdot L1 + (1-\alpha) \cdot L2 \rightarrow$ combination of both

💡 Trick: "L-ONE $\rightarrow$ LONE survivor (some weights become zero = feature selection)."

💡 Trick: "L-TWO $\rightarrow$ ALL shrink TWO-wards zero but nobody dies (all kept)."

✅ High-dimensional + irrelevant features $\rightarrow$ ALWAYS Lasso. Correlated features $\rightarrow$ Ridge.

✅ $\lambda \uparrow$ = more regularisation = more bias = less variance. $\lambda \downarrow$ = less regularisation.

Polynomial Basis Functions

Add x^2 , x^3 , x_1x_2 as new features $\rightarrow$ non-linear fit with linear algorithm

Still linear in PARAMETERS $\theta \rightarrow$ same gradient descent applies

Higher degree $\rightarrow$ lower bias, higher variance $\rightarrow$ overfits

💡 Trick: "Degree 1 = underfits curves. Degree 3-4 = sweet spot. Degree 10+ = wiggly overfit."

Module 4 — Linear Classification & Logistic Regression

★ Sigmoid Function

$\sigma(z) = 1 / (1 + e^{-z})$ where $z = \theta^T x = \theta_0 + \theta_1 x_1 + \dots$

Range: (0, 1) always. $\sigma(0)=0.5$. $\sigma(+\infty) \rightarrow 1$. $\sigma(-\infty) \rightarrow 0$.

Derivative: $\sigma'(z) = \sigma(z) \cdot (1 - \sigma(z))$

Decision: predict class 1 if $\sigma(z) \geq 0.5 \leftrightarrow z = \theta^T x \geq 0$

💡 Trick: "S-curve squashes any real number into (0,1). Linear Reg can't do this."

💡 Trick: Large weights $w \rightarrow$ STEEP sigmoid $\rightarrow$ near step function $\rightarrow$ overfit (memorises boundary).

★ Log-Loss (Cross-Entropy)

$J(\theta) = -(1/m) \sum [y_i \log(\sigma(\theta^T x_i)) + (1-y_i) \log(1-\sigma(\theta^T x_i))]$

When $y=1$ & $\hat{h} \approx 0 \rightarrow$ penalty = $-\log(0) \rightarrow \infty$ (correct wrong confident prediction)

When $y=1$ & $\hat{h} \approx 1 \rightarrow$ penalty = $-\log(1) \rightarrow 0$ (no penalty)

💡 Trick: "Log-loss is CONVEX with sigmoid. MSE with sigmoid is NOT convex (local minima)."

★ Gradient Descent for Logistic Regression

$\theta_j := \theta_j - \alpha \cdot (1/m) \sum (\sigma(\theta^T x_i) - y_i) \cdot x_{ij}$

⚠️ Looks IDENTICAL to Linear Regression update! Only $h\theta(x)$ differs.

Linear Reg: $h\theta(x) = \theta^T x$. Logistic Reg: $h\theta(x) = \sigma(\theta^T x)$.

💡 Trick: "Same update formula, different hypothesis function. Remember this — it's a common exam trick!"

★ Log-Odds Interpretation

$$\log[P(y=1|x) / P(y=0|x)] = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots$$

Positive $\theta_j \rightarrow$ increasing x_j raises probability of class 1

Large positive $\theta_j \neq$ guarantee of class 1 (other features also matter!)

💡 Trick: "Log-odds is LINEAR. Each θ_j = contribution of feature j to the log-odds of class 1."

✅ Exam trap: "Large θ_{CGPA} guarantees selection" $\rightarrow$ WRONG. All features contribute jointly.

Decision Boundary & Multiclass

Boundary: $\theta^T x = 0$ (where $\sigma(z) = 0.5$)

2D: $\theta_0 + \theta_1 x_1 + \theta_2 x_2 = 0 \rightarrow$ a straight LINE

Multiclass softmax: $P(y=k|x) = \exp(\theta_k^T x) / \sum_j \exp(\theta_j^T x)$

💡 Trick: "Logistic boundary = LINE in feature space. Non-linear boundary needs polynomial features first."

Module 5 — Decision Trees

★ Entropy Formula

$$H(S) = - \sum_i p_i \log_2(p_i)$$

$H = 0 \rightarrow$ PURE node (all same class). $H = 1 \rightarrow$ 50/50 split (max disorder, binary).

Quick values: $H(\frac{1}{2}, \frac{1}{2}) = 1.0$ $H(\frac{3}{4}, \frac{1}{4}) \approx 0.811$ $H(1, 0) = 0$ $H(\frac{1}{4}$ each, 4 classes) = 2.0

💡 Trick: "Entropy = surprise/disorder. 0 = no surprise (all same). Max = totally random."

💡 Trick: $\log_2(1/2) = -1$, $\log_2(1/4) = -2$, $\log_2(1/8) = -3$. Memorise these!

★ Information Gain Formula

$$IG(S, A) = H(S) - \sum_v (|S_v| / |S|) \cdot H(S_v)$$

Choose attribute $A^* = \operatorname{argmax} IG(S, A)$

$IG = 0 \rightarrow$ attribute provides NO information. $IG = H(S) \rightarrow$ perfect split.

💡 Trick: "IG = how much entropy DECREASES after splitting. Higher IG = better question."

💡 Trick: "Pure branches (all same class) $\rightarrow H=0$ after split $\rightarrow IG = H(S)$ = maximum possible."

✅ Exam: If you see a branch where all examples are same class $\rightarrow$ that branch is a LEAF immediately.

★ ID3 Algorithm Steps

Step 1: If all examples = same class $\rightarrow$ return LEAF with that class

Step 2: If no attributes left $\rightarrow$ return LEAF with MAJORITY class

Step 3: For each attribute $A \rightarrow$ compute $IG(S, A)$

Step 4: Select A^* with maximum IG $\rightarrow$ make it the node

Step 5: For each value v of A^* → create branch → recurse on subset S_v

💡 Trick: "ID3 = Greedy Top-Down. Always picks the BEST question at each step. Never backtracks."

Gini Impurity (CART Alternative) ▶

$$\text{Gini}(S) = 1 - \sum_i p_i^2$$

Gini = 0 → pure. Gini = 0.5 → 50/50 binary (max impurity).

Faster than entropy (no log computation). Used by scikit-learn CART by default.

💡 Trick: "Gini = 1 minus the sum of squares of probabilities. Entropy uses logs. Both measure impurity."

★ Overfitting & Pruning ▶

Problem: Deep tree → perfect train accuracy, poor test accuracy (memorised noise)

Pre-pruning: max_depth, min_samples_leaf, min_IG threshold → stop early

Post-pruning: Grow full tree → prune branches that hurt validation accuracy

MDL: minimise $L(\text{model}) + L(\text{data}|\text{model})$ → prefer simpler trees

💡 Trick: "Pre-pruning = stop early (don't grow). Post-pruning = trim after full growth."

ID3 vs C4.5 vs CART — Quick Compare ▶

ID3: Information Gain · Categorical only · No pruning · Multi-way splits

C4.5: Gain Ratio · Categorical + Continuous · Post-pruning · Multi-way splits

CART: Gini Impurity · Both types · Cost-complexity pruning · Binary splits only
· Regression too

💡 Trick: "ID3 = Introductory. C4.5 = Corrected (Gain Ratio fixes IG bias). CART = Commercial (scikit-learn)."

📄 One-Page Formula Summary — Print & Stick

Module	Most Important Formula	💡 Memory Trick	★ Exam Weight
M1 — Intro	T/E/P Definition. Types: Supervised / Unsupervised / RL	"Labels? → Supervised. Rewards? → RL. Neither? → Unsupervised."	~15%
M2 — Workflow	Z-score: $x' = (x - \mu) / \sigma$ · $F1 = 2PR / (P + R)$ · Error = Bias ² + Var + Noise	"Gap in train/val = Variance. Both high = Bias."	~25%
M3 — LinReg	$\theta = (X^T X)^{-1} X^T y$ · Ridge: $+\lambda \sum \theta^2$ · Lasso: $+\lambda \sum \theta $	"L-ONE kills features. L-TWO shrinks all."	~20%
M4 — LogReg	$\sigma(z) = 1 / (1 + e^{-z})$ · $J = -\sum [y \cdot \log(\hat{h}) + (1 - y) \cdot \log(1 - \hat{h})]$	"Large weights → steep sigmoid → overfit."	~25%
M5 — DTree	$H(S) = -\sum p \cdot \log_2 p$ · $IG = H(S) - \sum (S_v / S) \cdot H(S_v)$	"IG = how much entropy DROPS. Higher = better split."	~20%



Model Comparison — All 5 Modules

Side-by-side comparison of all models covered in scope

Model	Type	Decision Boundary	Overfits?	Interpretable?	Feature Scaling?	Loss Function	Best For
Linear Regression	Regression	Hyperplane (continuous output)	Yes (high-degree poly)	Yes (coefficients)	Yes	MSE	Continuous target, few features
Ridge Regression (L2)	Regression	Hyperplane	Less than LR	Yes	Yes	$MSE + \lambda \sum \theta^2$	Many correlated features, keep all
Lasso Regression (L1)	Regression	Hyperplane	Less than LR	Yes (sparse)	Yes	$MSE + \lambda \sum \theta $	High-dim, sparse, irrelevant features
Logistic Regression	Classification	Linear (smooth)	Moderate	Yes (log-odds)	Yes	Log-loss	Binary/multi-class, probabilistic output
Decision Tree (ID3/CART)	Classification/Regression	Axis-aligned, piecewise	High	Very high	No	Entropy/Gini	Non-linear, mixed features, explainability
KNN (Module 6)	Both	Non-linear, local	High (small k)	Moderate	Yes	Distance metric	Local patterns, non-parametric

vs Lasso vs Ridge — Decision Guide

- **Use Lasso when:** High-dimensional data, many irrelevant features, want automatic feature selection, need sparse model, 10–15 of 500 features are important
- **Use Ridge when:** All features are somewhat relevant, many correlated features, don't want to discard any feature entirely
- **Use Elastic Net when:** Both sparsity and stability are needed

vs Logistic Regression vs Decision Tree

- **Use Logistic Regression when:** Linear separability (approximately), need probabilistic output, need stable predictions, interpretable coefficients needed
- **Use Decision Tree when:** Non-linear relationships, mixed feature types, need human-readable rules, no feature scaling possible

vs Batch GD vs SGD vs Mini-batch

- **Batch GD:** Stable, smooth convergence, slow for large data
- **SGD:** Fast, noisy, good for streaming/online learning
- **Mini-batch:** Best balance → used in deep learning (batch size 32–256)



Last-Hour Exam Tips

High-frequency patterns, common traps, and critical formulas to remember

🔥 Most Repeated Questions (2021–2026)

- Tom Mitchell T/E/P identification (EVERY year)
- Type of ML for 3–4 scenarios (EVERY year)
- Data quality issues with 6+ fixes (EVERY year)
- L1 vs L2 regularisation — choose for scenario (EVERY year)
- Entropy + IG calculation for decision tree (EVERY year)
- Bias-variance diagnosis from train/val error (EVERY year)
- Confusion matrix + Precision/Recall/F1 (EVERY year)

⚠️ Common Exam Traps

- **Accuracy trap:** 95% accuracy with 95% majority class → always check for imbalance
- **λ direction:** High bias → DECREASE λ (not increase). High variance → INCREASE λ .
- **Log₂ vs ln:** Decision trees always use log₂ for entropy in bits
- **Lasso for high-dim:** Always choose Lasso (not Ridge) for high-dimensional sparse data
- **Test set leakage:** Any use of test data during training invalidates evaluation
- **Large weights → overfit:** Large $|\theta|$ in logistic regression → steep sigmoid → overfit

📐 Must-Know Calculations

Entropy: $H = -\sum p \log_2 p$
Pure node: $H=0$ (one class only) Binary
50/50: $H=1$ (maximum)
Precision = $TP / (TP+FP)$
← "Of what I said +, how many were +"
Recall = $TP / (TP+FN)$ ←
"Of actual +, how many did I catch?" $F1 = 2PR / (P+R)$ Z-score:
 $(x-\mu) / \sigma$ Min-Max:
 $(x-\min) / (\max-\min)$
Sigmoid: $\sigma(z) = 1 / (1+e^{-z})$ $\sigma(0)=0.5$,
 $\sigma(\text{large}+) \approx 1$,
 $\sigma(\text{large}-) \approx 0$

✅ 6-Hour Study Order

- **Hour 1:** Module 1 + Module 2 summaries (all questions are definition-based)
- **Hour 2:** Module 2 calculations (Z-score, Min-Max, confusion matrix, metrics)
- **Hour 3:** Module 3 (Gradient descent update, L1 vs L2, normal equation)
- **Hour 4:** Module 4 (Sigmoid, logistic regression pipeline, log-odds)
- **Hour 5:** Module 5 (Entropy calculation, full IG worked example, ID3)
- **Hour 6:** Skim ALL question papers; re-do starred ★ questions from memory

📊 Quick Calculation Tips

- $\log_2(1/2) = -1$; $\log_2(1/4) = -2$;
 $\log_2(1/8) = -3$
- $\log_2(3/4) \approx -0.415$; $\log_2(1/4) = -2$
- $H(50:50) = 1.0$ bit; $H(75:25) \approx 0.811$;
 $H(100:0) = 0$
- $\sigma(0) = 0.5$; $\sigma(1) \approx 0.731$; $\sigma(2) \approx 0.880$;
 $\sigma(-1) \approx 0.269$
- $e^{-1} \approx 0.368$; $e^{-2} \approx 0.135$

📊 Exam Weight Estimate

- **Module 1:** ~15% (3–4 marks, always Q1)
- **Module 2:** ~25% (5–8 marks, data quality + metrics)
- **Module 3:** ~20% (4–6 marks, gradient descent + regularisation)
- **Module 4:** ~25% (5–8 marks, logistic regression pipeline)
- **Module 5:** ~20% (4–8 marks, entropy/IG tree building)